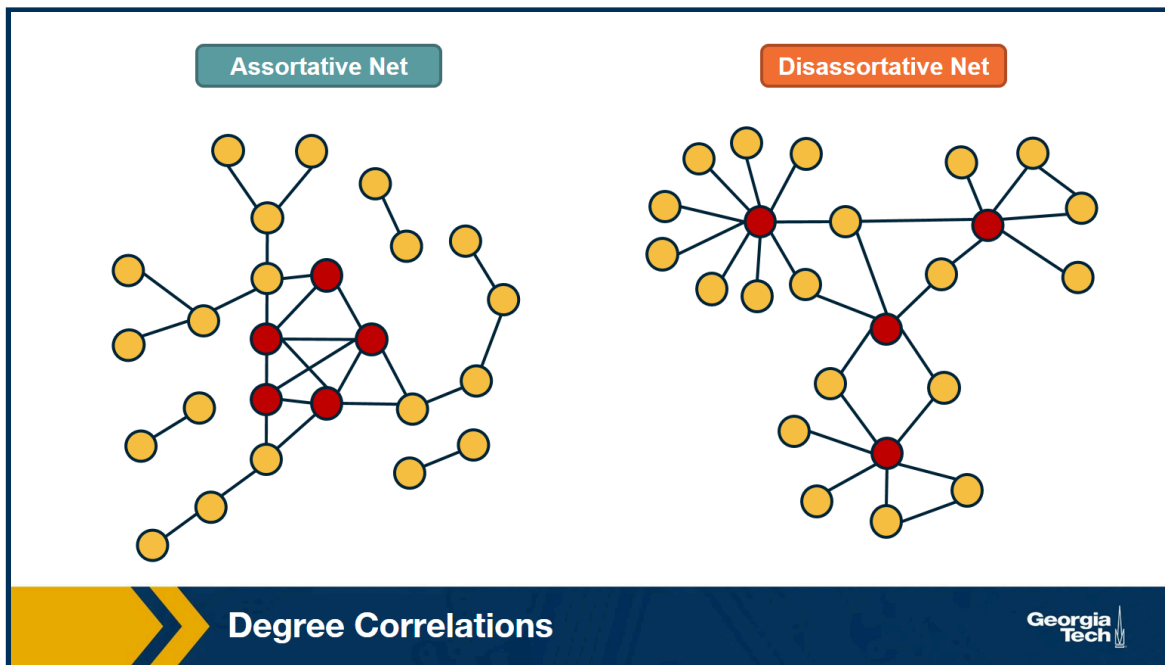


L3: Degree Correlations



We assumed throughout this lesson that the degree of a node does not depend on the degree of its neighbors. In other words, **we assumed that there are no degree correlations**.

Mathematically, if nodes u and v are connected, we have assumed that;

$$\text{Prob}[\text{degree}(u) = k \mid \text{degree}(v) = k'] =$$

$$\text{Prob}[\text{degree}(u) = k \mid u \text{ connects to another node}] = q_k = p_k \cdot \frac{k}{\bar{k}}$$

Note: this probability does not depend on the degree k' of neighbor v . Such networks are referred to as **neutral**.

In general, however, there are correlations between the degrees of neighboring nodes, and they are described by the conditional probability distribution:

$$P[k' | k] = \text{Prob}[\text{a neighbor of a } k\text{-degree node has degree } k']$$

The expected value of this distribution is referred to as the **average nearest-neighbor degree** $k_{nn}(k)$ of degree- k nodes:

$$k_{nn}(k) = \sum_{k'} k' \cdot P(k' | k)$$

In a neutral network, we have already derived that $k_{nn}(k)$ is independent of k (recall that we derived

$$k_{nn}(k) = \bar{k} + \frac{\sigma_k^2}{\bar{k}} = \bar{k}_{nn}$$

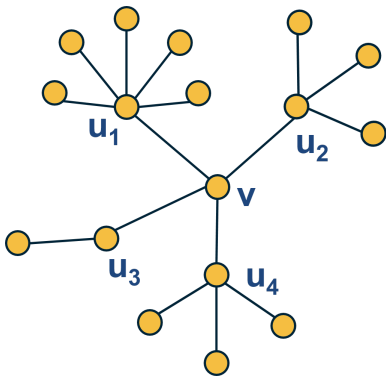
In most real networks, $k_{nn}(k)$ depends on k and it shows an increasing or decreasing trend with k .

The network at the left shows an example in which small-degree nodes tend to connect with other small-degree nodes (*and similarly for high-degree nodes*).

The network at the right shows an example of a network in which small-degree nodes tend to connect to high-degree nodes.

Degree Correlation Example

Here is an example: what is the average nearest neighbor degree of node v in this network?



$$\overline{k_{nn}}(v) = \frac{1}{k(v)} \sum_{i=1}^{k(v)} k(u_i) = \frac{6+4+2+4}{4} = 4$$

To calculate $k_{nn}(k)$ we compute the average value of $k_{nn}(v)$ for all nodes v with $k(v) = x$.

Then, we plot $k_{nn}(k)$ versus k , and examine whether that plot shows a statistically significant positive or negative trend.